

Bessel Beam Optics for Interrogation of 1-Meter Vertical Strontium Optical Lattice Clock

Physics Innovators Initiative (Pi²) Summer 2026 Cohort

Leigh Zaman,^{1,*} Anke Stoeltzel,^{1,†} Shuo Ma,¹ and Shimon Kolkowitz¹

¹*Kolkowitz Group, Department of Physics, University of California, Berkeley, CA*

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We designed, built, and characterized a three-axicon system that forms a zeroth-order Bessel beam from 1 in. optics and evaluated four configurations for the interrogation beam in a strontium optical lattice clock. Using a purpose-built motorized three-axis imaging system, we profiled the beam over 2 m of propagation. All four produced Bessel cores with mean diameter within 8% of the ideal J_0^2 value, but none met the required 300 μm FWHM. The widest measured central lobe was 125.6 μm . A quasi-discrete Hankel transform (QDHT) model with no free parameters predicted Bessel-zone start within 10% and length within 8% for two of three modeled configurations. A geometric ray model predicted the start comparably well but overestimated zone length by 42–89%. The longest measured zone was $>0.83\text{ m}$ against a 1 m target, while on-axis intensity varied by 48–63% peak-to-peak in all measured zones. The 1 in. aperture limits axicon separation distance, which limits zone length and creates periodic ripples in intensity along the propagation axis. None of the four configurations met all design requirements.

I. BACKGROUND AND AIMS

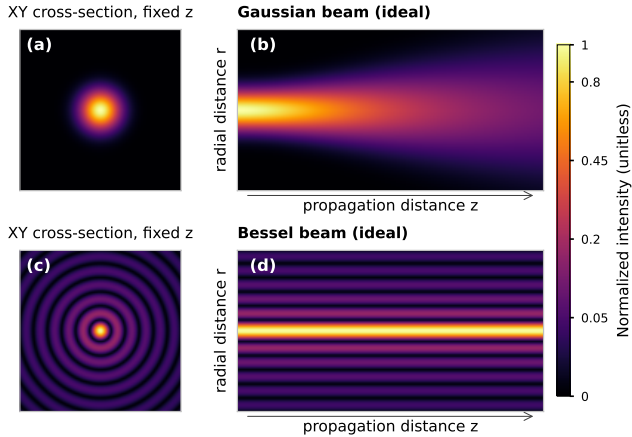


FIG. 1. Comparison of an ideal Gaussian beam and zeroth-order Bessel beam. (a) Transverse intensity profile of a Gaussian beam at its waist. (b) Transverse intensity profile $I(r) \propto J_0^2(k_r \cdot r)$ of an ideal Bessel beam, showing the central lobe and surrounding concentric rings. (c) Longitudinal intensity profile of a Gaussian beam, showing how the beam's width grows (diverges) as it propagates. (d) The Bessel beam's central lobe stays constant/uniform.

Optical lattice clocks are precise enough to resolve the gravitational redshift at tabletop scale. The first-generation strontium clock in our lab resolved the redshift across five atom ensembles spanning a 1 cm vertical path [1]. The second-generation clock extends this

to three ensembles across 1 m. A vertical optical lattice traps the strontium atoms, and a co-propagating 698 nm clock laser interrogates (probes) them. Since the redshift is proportional to height, intensity variations along the interrogation path introduce a systematic error that must be characterized. The clock beam would ideally be as uniform as possible across over a meter of propagation, which is not achievable with a Gaussian beam and 1-2 in. lens elements (see Figure 1).

A Bessel beam is a promising candidate because it has a non-diffracting profile over long propagation distances [2–4]. This study addresses one primary question: can a three-axicon system built from 1 in. optics produce a Bessel zone that meets the following requirements:

1. Bessel zone length of at least 1 m.
2. Central lobe large enough to contain a cloud of strontium atoms (cross-sectional FWHM at least 300 μm in diameter).
3. Uniform on-axis intensity profile.
4. Beam within the Bessel region fits within 1/2 in. mirror aperture (mirror 1 in Figure 2, entrance to science chamber).
5. Annulus after Bessel region is larger than 1/2 in. to minimize reflections off mirror 2 in Figure 2.

We tested this by building four candidate configurations (Table I) and comparing the measured Bessel zone of each against models described in section II B and section II C.

* leighzaman@berkeley.edu

† Mentor

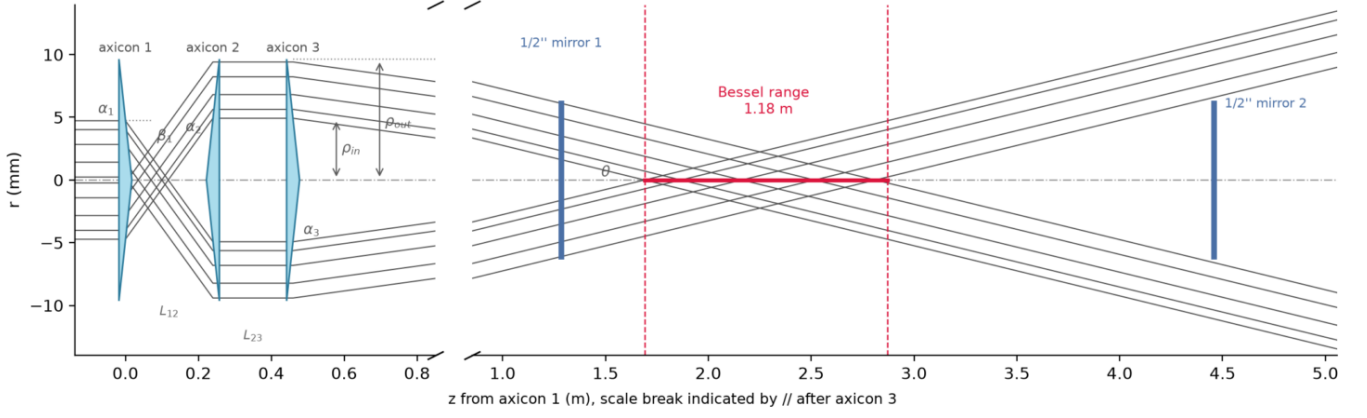


FIG. 2. Geometric model of the three-axicon Bessel beam system, shown for candidate #2 (Table I). Axicon 1 ($\alpha_1 = 5^\circ$) converts the Gaussian input beam into a cone of rays deflected by β_1 . Axicon 2 ($\alpha_2 = 5^\circ$) collimates them into an annulus of inner radius ρ_{in} and outer radius ρ_{out} . Axicon 3 ($\alpha_3 = 0.5^\circ$) deflects the annulus rays by θ . The region where the rays overlay on axis (red) is the geometric Bessel range (shown here as 1.18 m). L_{12} and L_{23} are the axicon separations. The first 1/2 in. mirror steers the beam into the science chamber, and the second 1/2 in. mirror must be placed in the void zone to minimize stray reflections.

II. EXPERIMENTAL AND THEORETICAL APPROACHES

A. Bessel Beams and Bessel-Gauss Beams

On the plane transverse to the propagation direction, an ideal Bessel beam (BB) has an invariant intensity distribution described by a Bessel function of the first kind, J_0 . This property led to the description of BBs as non-diffracting beams, since the transverse intensity profile does not change as the beam propagates through space [2]. The ideal Bessel beam is non-diffracting over a theoretically infinite range. The transverse intensity distribution of an ideal Bessel beam has the form:

$$I(r) \propto J_0^2(k_r \cdot r), \quad (1)$$

where k_r is the transverse wavevector satisfying $k_r^2 + k_z^2 = k^2 = (2\pi/\lambda)^2$ and r is the radial distance from the propagation axis.

Passing a Gaussian beam through an axicon (another name for a conical prism) produces a Bessel-Gauss beam by converting the Gaussian input into a conical wavefront that interferes constructively along the propagation axis [4]. An axicon of base angle α and refractive index n deflects the Gaussian beam toward the axis by $\beta = \arcsin(n \sin \alpha) - \alpha$, creating a Bessel-Gauss beam of the zeroth order. The Bessel-Gauss beam has a transverse intensity profile described by Equation 1, where $k_r = (2\pi/\lambda) \sin \beta$ is governed by the axicon deflection angle β and the wavelength λ of the input Gaussian beam.

B. Geometric / Ray Model

We based the design on the collimating three-axicon zoom system described in Schnieder *et al.* [5]. Figure 2 shows the geometric model of our system and defines the ray angles and axicon separations. Because the annulus

leaving axicon 1 keeps expanding as it propagates, the 1 in. aperture of axicon 2 sets an upper limit on L_{12} . Within that limit, increasing L_{12} suppresses the on-axis intensity ripple (Figure 3).

C. Wave Optics / QDHT Model

The geometric model predicts where rays overlap, but not the intensity within the Bessel zone. To get a sense of the idealized intensity distribution, we constructed a wave-optics simulation. We first attempted the analysis in Zemax software, whose physical optics propagation uses a fast Fourier transform (FFT) on a square Cartesian grid [6]. We discarded this approach out of concern that design decisions would be governed by non-physical artifacts of the square sampling grid.

Since the idealized optical system is rotationally symmetric about the propagation axis, we were able to model the field propagation with a quasi-discrete Hankel transform (also known as a Fourier-Bessel transform) described by Yu *et al.* [7].

Our QDHT model implementation accepts only measured quantities as inputs: the Gaussian input beam waist w_0 , the axicon base angles α_1 , α_2 , and α_3 , the axicon separations L_{12} and L_{23} , and the wavelength λ . The model has no free parameters fit to our data. The QDHT models in Figure 6 are sampled at $N = 4096$ points in the radial direction over a domain padded to $2.5\times$ the Gaussian input beam radius.

We implemented both geometric and QDHT models in an open-source Python simulator with an interactive web interface¹. Parameter sweeps over the axicon angles, separations, and input beam diameter are collected in

section A.

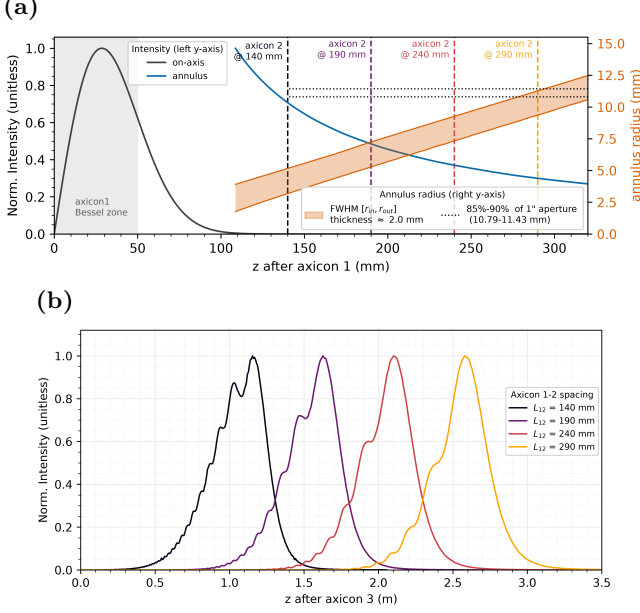


FIG. 3. Effect of axicon 1-2 separation L_{12} in optics configuration #1 (see Table I), simulated with QDHT. (a) Propagation between axicons 1 and 2: after axicon 1 Bessel zone (shaded in grey), the beam becomes an expanding annulus with approximately 2 mm FWHM thickness (shaded orange region). L_{12} sets the annulus radius arriving at axicon 2 (annulus radius shown on right y-axis). The 85-90% aperture limit of 1 in. optics (black dotted line) determines the upper limit on L_{12} separation. (b) Comparison of on-axis intensity after axicon 3 for $L_{12} = 140, 190, 240, 290$ mm. Increasing the L_{12} separation suppresses the on-axis intensity ripples.

D. Experimental Setup

We coupled an adjustable-focus laser diode (Thorlabs CPS650F, nominal $\lambda = 650$ nm) to a single-mode fiber with a fiber optic coupler (Schafter+Kirchhoff 60FC-L-4-M25-24), producing a Gaussian input beam with $1/e^2$ diameter $D_{\text{in}} = 4.59$ mm at axicon 1 and 1.2 mW of power after the coupler.

We mounted three UV fused silica axicons (Thorlabs AX255-A and AX2505-A; $n \approx 1.4565$ at 650 nm, AR-coated for 350–700 nm; $\alpha_1 = \alpha_2 = 5^\circ$, $\alpha_3 = 0.5^\circ$) in XY translation mounts and measured the separations L_{12} and L_{23} with a stainless steel ruler. We aligned the system by centering the annulus on each successive optic (“beam walking”), using the beam profiler gantry as the alignment tool. The four candidate configurations we evaluated are listed in Table I.

We characterized the Bessel zone by imaging the beam through an OD=4.0 filter directly onto a monochrome CMOS sensor (FLIR BFS-PGE-31S4M-C, 2048×1536

pixels, $3.45 \mu\text{m}$ pixel size) mounted on a motorized three-axis CNC gantry built for this project, which allowed us to sample the beam profile along 2 m of propagation with repeatable sub-millimeter positioning. The gantry has 130 mm of travel along the propagation direction and 100 mm of travel in the transverse XY plane. Because the z travel is shorter than the region of interest, we repositioned the gantry by hand along the optical table to cover the full 2 m propagation path. Each placement is registered against a sheet of 1 cm graph paper fixed to the table (Figure 4), so the measurement position is repeatable between runs. At each plane the acquisition software locates the beam and sets the exposure time automatically, so that the brightest pixel in the plane falls below a saturation threshold. It also collects dark frames for background subtraction. If the beam diameter is larger than the camera sensor, the profiler software collects a grid of images and stitches them into a single composite image.

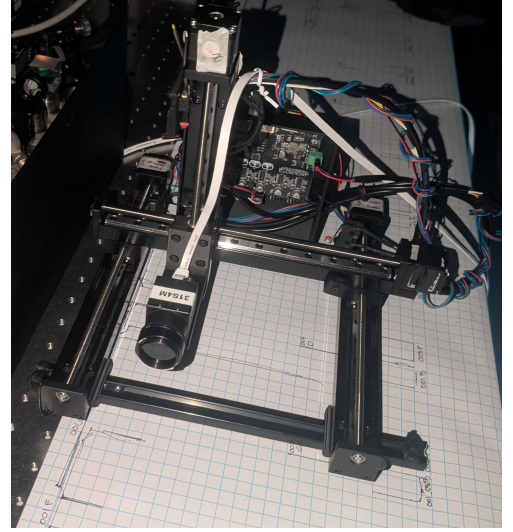


FIG. 4. We built a motorized three-axis CNC gantry for this project. The CMOS camera rides on the moving stage and images the beam through an OD=4.0 filter. The 1 cm graph paper beneath the gantry frame registers each hand-placement of the gantry along the optical table.

E. Metrics

From each plane, we crop the image to the central Bessel lobe. We report intensity as pixel counts / exposure time, so planes recorded with different exposures are comparable. We extract the on-axis intensity $I(r = 0, z)$ by fitting a J_0^2 function to the data (see Figure 5), where the amplitude A represents the peak intensity of the plane. From the fitted k_r wavenumber, we calculate the central lobe diameter $\varnothing$ and mean radial FWHM $\langle \text{FWHM} \rangle_r$. The measured Bessel zone Z_R is the interval between the outermost points at which $I(r = 0, z)$ falls to 50% of its maximum, and ΔZ_R is the length of that interval. The ripple metric is the peak-to-peak variation

¹ <https://bessel-beam.streamlit.app/>

TABLE I. Optics configuration candidates. All configurations use 1 in. (25.4 mm) axicons with base angles $\alpha_1 = \alpha_2 = 5^\circ$ and $\alpha_3 = 0.5^\circ$. L_{12} and L_{23} are the axicon 1–2 and axicon 2–3 separations, D_{in} is the $1/e^2$ diameter of the Gaussian input beam at axicon 1, and $\lambda = 650$ nm is the wavelength of the input laser diode.

#	Description	L_{12} (mm)	L_{23} (mm)	D_{in} (mm)	2 $\times$ telescope	Dataset (# of images)
1	Baseline	190	50	4.59	—	Jul 22-23 (1,202)
2	Gaussian input beam magnified 2 $\times$ vs. baseline (#1)	240	220	9.44	input ^a	Jul 29-30 (1,013)
3	Hollow Gaussian beam diameter reduced vs. baseline (#1)	175	415	4.59	—	Aug 6 (411)
4	Bessel zone magnified 2 $\times$ vs. #3	175	415	4.59	Bessel zone ^b	Aug 6 (241)

^a 2 $\times$ magnification telescope before axicon 1, expanding the Gaussian input beam. Telescope constructed with 1 in. lenses: $f_1 = -50$ mm (plano-concave), $f_2 = 100$ mm (plano-convex) mounted in a 1 in. optical cage.

^b 2 $\times$ magnification telescope placed within the Bessel zone at $z = 120$ cm after axicon 3. Telescope constructed with 2 in. lenses: $f_1 = 100$ mm, $f_2 = 200$ mm, mounted directly to the optics breadboard table.

of $I(r = 0, z)$ within the measured Bessel zone, normalized to the mean on-axis intensity. The beam profiling software, analysis notebooks, and mechanical designs are open source².

III. RESULTS

None of the four configurations using 1 in. axicons described in Table I met the design requirements described in section I. Three of the five failed on measured data. The longest Bessel zone was >0.83 m against a 1 m target. The largest central lobe was $125.6 \mu\text{m}$ FWHM against the $300 \mu\text{m}$ needed to contain the atom cloud. The on-axis intensity varied by 48-63% of the mean, so it was not uniform. Requirements 4 and 5 (tests concerned with the two science-chamber mirrors) are deferred to future work (section IV A).

The measured Bessel zone for each configuration is summarized in Table II and Figure 6, along with the modeled geometric and QDHT ranges. We evaluated four candidates, described in Table I. The QDHT simulation consistently predicted the starting point of the Bessel range for three candidates within 10% agreement. In two of the three modeled candidates, the Bessel zone length was within 8% agreement with the QDHT simulation. The geometric model consistently predicted the start of the Bessel zone within 3.6% to 5.6% agreement but overestimated the Bessel zone length by +42% to +89%.

Candidate #1 was the baseline configuration, and it established how closely the models describe a well-aligned system. The measured zone length fell within 1 cm of the QDHT prediction, the closest agreement of any candidate, and the core diameter and FWHM tracked the ideal J_0^2 values to within 2% at every plane inside the Bessel zone.

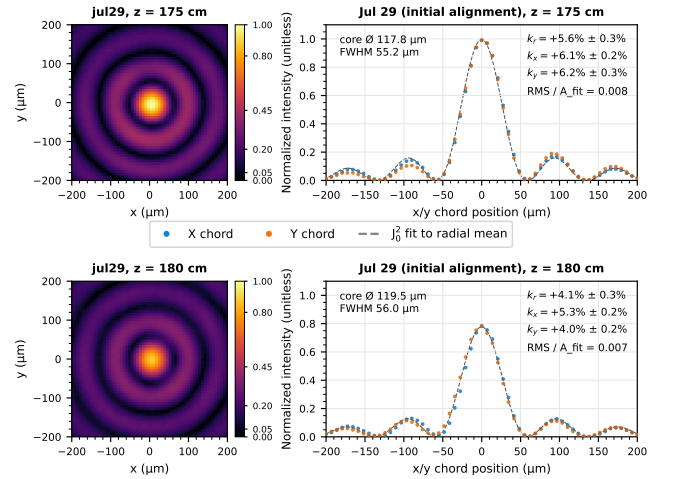


FIG. 5. Data from the Bessel zone of optics configuration #2 in Table I. (a) Cross-section image of beam profile at $z = 175$ cm. Image data is cropped to $\pm 200 \mu\text{m}$ of the central Bessel lobe, colored by intensity (pixel counts / exposure time) and normalized. (b) Measured intensity sampled on a 2×2 pixel grid ($6.9 \times 6.9 \mu\text{m}$) and fit to a J_0^2 function (black line) to extract the core diameter and FWHM. (c) $z = 180$ cm cross-section, intensity normalized against the peak intensity measurement at $z = 175$ cm shown in (a). (d) J_0^2 fit to the $z = 180$ cm cross-section.

Candidate #2 used a 2 $\times$ magnification telescope before axicon 1 to expand the Gaussian input beam. Expanding the input beam substantially increased the Bessel zone length ($\Delta Z_R > 0.83$ m), but at the cost of worse on-axis intensity ripples. The spacing $L_{12} = 240$ mm was the maximum allowed by the 1 in. aperture of axicon 2, so this configuration represents the longest Bessel zone reachable with 1 in. optics. Space limitations on the optics table prevented full characterization, so Z_R and ΔZ_R are lower bounds. Core diameter and FWHM tracked the ideal values to within 5%.

Candidate #3 repeated candidate #1 with the axicon spacings changed (Table I) to shrink the hollow Gaussian beam reaching axicon 3. It produced the shortest zone

² https://github.com/leigh-johnson/bessel_beam_profiler

TABLE II. Measured and modeled Bessel zone for the optics configuration candidates described in Table I. All z values are measured along the propagation direction from the face of axicon 3. Z_R is the range of the Bessel region and ΔZ_R its length ($\pm$ imaging system z -step size). The mean core diameter $\varnothing$ and core FWHM are averages of the per-plane J_0^2 fits within the measured Bessel zone (deviation from ideal value in parentheses). The geometric range is calculated by the method described in section II B; the QDHT range is the FWHM of the modeled on-axis intensity $I(0, z)$ described in section II C. Parenthetical percentages on model rows give the deviation from the measured value, and are upper limits where the measurement is a lower bound. No simulation/model was developed for configuration #4. Ripple (%) is the ratio of the peak-to-peak variation in the on-axis intensity $I(0, z)$ to the mean on-axis intensity within the measured Bessel zone. Peak intensity (counts/ μ s) is the maximum measured on-axis intensity within the measured Bessel zone, used to normalize the measured on-axis intensity in Figure 6.

Config	Method	Z_R (m)	ΔZ_R (m)	Core $\varnothing$ (μ m)	$\langle \text{FWHM} \rangle_r$ (μ m)	Ripple (%)	Peak (counts/ μ s)
#1	measured	$[1.29, 1.68] \pm 0.01$	0.38 ± 0.01	126.4 (+1.7%)	59.2	48	0.723
	geometric	$[1.34 (+3.6\%), 1.91 (+13\%)]$	$0.57 (+49\%)$				
	QDHT	$[1.40 (+8.2\%), 1.77 (+4.9\%)]$	$0.37 (-3\%)$				
#2 ^a	measured	$[1.18, >2.01]$	>0.83	118.2 (-5.0%)	55.4	63	0.590
	geometric	$[1.23 (+4.2\%), 2.41 (<+20\%)]$	$1.18 (<+42\%)$				
	QDHT	$[1.29 (+9.3\%), 2.19 (<+9\%)]$	$0.90 (<+8\%)$				
#3 ^b	measured	$[1.12, 1.42] \pm 0.01$	0.30 ± 0.01	125.0 (+0.5%)	58.5	63	2.003
	geometric	$[1.18 (+5.6\%), 1.76 (+24\%)]$	$0.57 (+89\%)$				
	QDHT	$[1.21 (+8.3\%), 1.62 (+14\%)]$	$0.41 (+36\%)$				
#4 ^c	measured	$[0.47, 0.79] \pm 0.01$	0.32 ± 0.01	268.1 (+7.8%)	125.6	50	0.323

^a Experimental setup did not allow for measurement of the Bessel zone beyond 2.0 m. The measured Z_R and ΔZ_R should be interpreted as lower bounds. See section IV A for discussion of how to extend the measurement range.

^b Alignment of axicons 2 and 3 was sub-optimal, resulting in a reduced ΔZ_R . See Figure 14 for a comparison of candidate #1 and #3 alignment.

^c A limited dataset was collected for configuration #4. The z axis measurements are taken after the telescope output lens instead of after axicon 3.

among the candidates. The core diameter and FWHM tracked the ideal values to within 1%. Unfortunately, this comparison is confounded by sub-optimal alignment of axicons 2 and 3, which would contribute to a reduced ΔZ_R (Figure 14). We cannot separate the effect of the spacing change from the effect of the misalignment, and this candidate also showed the largest QDHT disagreement in zone length.

Candidate #4 repeated the setup of Candidate #3, with a $2\times$ magnification telescope placed within the Bessel zone at $z = 120$ cm after axicon 3. The telescope doubled the transverse core size as expected, but, surprisingly, left the Bessel zone length nearly unchanged ($\Delta Z_R = 0.32 \pm 0.01$ m vs. 0.30 ± 0.01 m for #3) rather than stretching it by the expected longitudinal magnification. We examine this result in Sec. III C.

A. Ripples in On-Axis Intensity

The ripple % metric is defined in section II E. The on-axis intensity ripples varied by 48-63% of the mean within every candidate's Bessel zone (see Figure 6), which ruled out all four under the uniformity requirement.

In early design/modeling stages, we were unsure if the on-axis intensity ripples were a numerical artifact of the simulation sampling/aliasing. However, the ripples persisted across a wide range of sampling parameters. Initial measurements with a manual translation stage (see figure TODO) were too coarse to resolve them, but finer scans with the motorized gantry revealed on-axis intensity modulations in agreement with those predicted ana-

lytically by Rao [8] and reproduced by the QDHT simulations.

B. Axicon Alignment Drift

To test repeatability, we re-measured three z positions of candidate #2's Bessel zone five days after the original scan (Figure 5 vs. Figure 7). We reproduced the intensity ripples. The fitted core diameters agreed within a few percent, but the peak intensity dropped from 0.48 to 0.33 counts/ μ s, suggesting the drift did not occur uniformly. In Figure 7, the X-chord (blue dots) shifted more than the Y-chord (orange dots), resulting in a slight astigmatism. Fixing it required systematically adjusting the horizontal axes of all XY translation mounts to locate the source.

C. Effect of $M=2\times$ Telescope

Candidate #4 placed a $2\times$ magnification telescope inside the Bessel zone, $z = 120$ cm after axicon 3 (Table I), to understand the effect of magnification on the Bessel beam. The mean core diameter of $268.1 \mu\text{m}$ is within 7.8% of twice the ideal core size, confirming the expected $2\times$ transverse magnification in the XY plane. This behavior was similar to the effect on a Gaussian beam, where an $M = 2$ telescope doubles the waist of the Gaussian beam.

The peak intensity dropped to 0.32 counts/ μ s, compared to 2.00 for the identical optics without the telescope (candidate #3). Spreading the same power over a $4\times$ larger area ($M^2 = 4$) alone would predict 0.50 counts/ μ s. The remaining loss would only be consistent

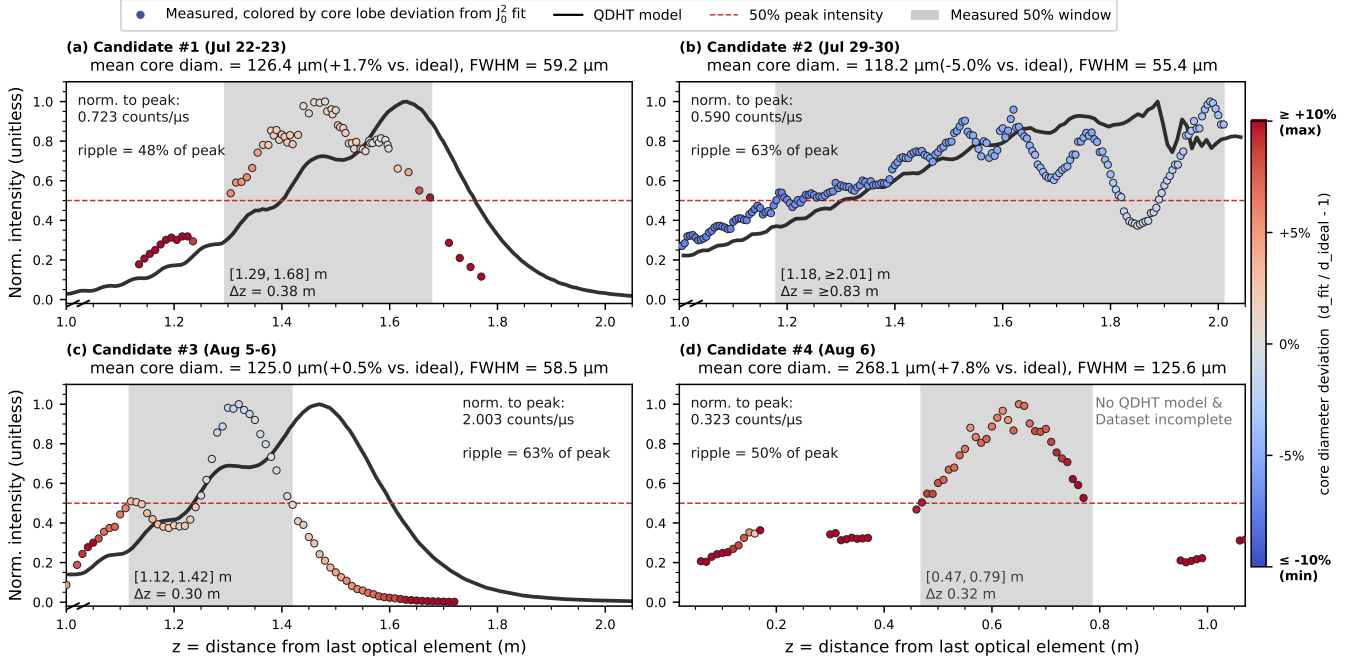


FIG. 6. Measured on-axis intensity $I(0, z)$ along the propagation direction for candidates (a) #1, (b) #2, (c) #3, and (d) #4 in Table I. Each panel normalized to its own peak intensity rate (counts/μs). z is measured from the last optical element. Points are J_0^2 amplitudes fit to cropped cross-section images of the plane transverse to the propagation direction. Points are colored by the deviation of the fitted central/core lobe diameter from the ideal value. The solid black line is the QDHT model, dashed red line indicates 50% of the measured peak, and the grey shaded region is the measured Bessel zone Z_R between the outermost 50% crossings. The ripple metric is the peak-to-peak variation of the on-axis intensity within the measured Bessel zone. Candidate #2's Bessel zone was not fully characterized due to space limitations on the optics table, so the measured Z_R and ΔZ_R should be interpreted as lower bounds. No QDHT model was developed for candidate #4, which placed a 2× magnification telescope inside the Bessel zone.

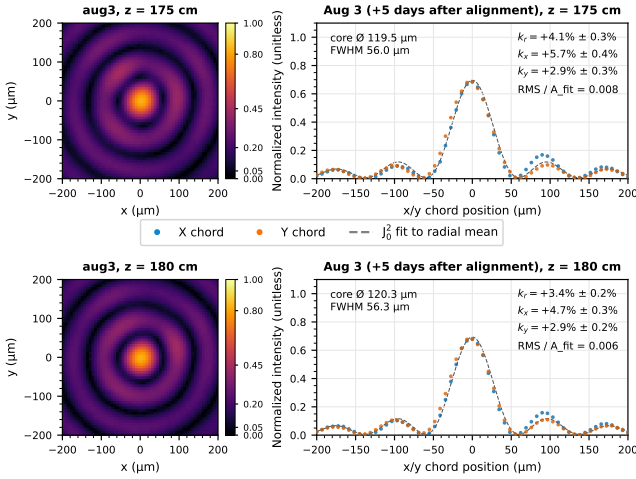


FIG. 7. TODO caption about alignment drift.

with ≈ 0.7 counts/μs ($\approx 35\%$ of the incident power) lost at the telescope lenses, so the coating and alignment of this configuration should be re-checked.

Candidate #4's dataset is the smallest we collected (241 images, Table I), it has a gap near $z \approx 0.2$ m,

and its z axis is referenced to the telescope output lens rather than to axicon 3. Subject to those limitations, we did not observe the expected longitudinal magnification of the Bessel zone. The measured length was $\Delta Z_R = 0.32 \pm 0.01$ m, against 0.30 ± 0.01 m for candidate #3 without the telescope. The missing planes near $z \approx 0.2$ m may contain a ripple peak that would extend the measured zone, in which case the magnification is present but unresolved. A re-scan of this configuration is needed to confirm the zone length scales as expected when a telescope acts on an already-formed Bessel zone.

IV. CONCLUSION AND PERSPECTIVE

We designed, built, and characterized a three-axicon Bessel beam system and evaluated four candidates for the interrogation beam in a 1 m vertical strontium optical lattice clock. Every configuration we tested produced a Bessel region with an identifiable core lobe in the cross-section. The measured core lobe diameters tracked the ideal J_0^2 values to within 8%. The QDHT wave-optics model predicted the start of the Bessel zone within 10% and, where we could measure it cleanly, the zone length

within 8%.

The tested three-axicon system, built from the present 1 in. optics, did not meet the clock-interrogation requirements. The longest measured zone (>0.83 m) fell short of the 1 m target. The central lobe reached at most $125.6\text{ }\mu\text{m}$ FWHM, well under the $300\text{ }\mu\text{m}$ specified atom cloud diameter. The on-axis intensity ripple of 48-63% far exceeded any acceptable uniformity threshold for clock interrogation.

Another open issue is day-to-day alignment drift, which needs characterization before any configuration is adopted. These results motivate the future work below, as well as the tooling we built to reduce the cost of every iteration that follows (automated beam profiler, QDHT simulator, axicon alignment tooling).

A. Future Work

Several open questions remain. First, we have not measured how the first 1/2 in. steering mirror (mirror 1 in Figure 2) affects the beam profile. The mirror sits upstream of the Bessel zone, and diffraction from its edges could distort the structure of the beam before it reaches the atoms. Second, axicons with 2 in. diameter have been ordered to raise the cap on L_{12} that the 1 in. aperture imposes. Third, day-to-day alignment drift limited the repeatability of our measurements, so the beam alignment should be monitored continuously. The au-

tomated alignment tool we built³ could be run between measurements to detect drift and alert when remediation is required. The alignment tool could also be extended to drive motorized translation mounts, ensuring the axicon lenses are always aligned to a certain tolerance. Ultimately, the final system will interrogate strontium atoms at 698 nm rather than the 650 nm of our test laser diode. The selected configuration will likely need to be re-characterized with the clock laser, since the transverse wavenumber k_r and Bessel zone geometry scale with wavelength.

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³ https://github.com/leigh-johnson/bessel_beam_profiler/blob/main/docs/axicon_alignment.md

Appendix A: QDHT Model Parameter Sweeps

The QDHT model described in section II C was used to sweep the axicon angles, separations, and Gaussian input beam diameter. The results are shown in Figure 8, Figure 9, Figure 10, and Figure 11. The model predicts that the Bessel zone length is most sensitive to axicon 3 angle α_3 and the Gaussian input beam diameter D . The Bessel core diameter is set by axicon 3 angle α_3 (and if applicable, any telescope placed in the Bessel region). The on-axis uniformity is governed by the quality of the Hollow Gaussian beam, which is set by the axicon 1-2 separation L_{12} .

All plots are shown for a fixed input beam diameter $D = 4.59$ mm at wavelength $\lambda = 650$ nm, unless otherwise noted.

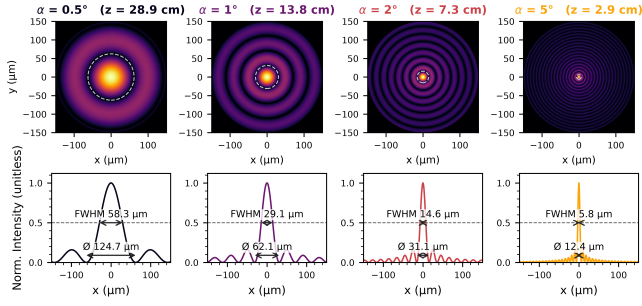


FIG. 8. QDHT model simulation of axicon with base angle α and Bessel core diameter ϕ for a single axicon. Core diameter is shown encircled by the dashed line in the rendered 2D cross-section, and FWHM is shown to scale on the 1D plot. All cross-sections are shown at the predicted peak intensity point ($z = 28.9, 13.8, 7.3$, and 2.9 cm respectively). Fixed input beam diameter $D = 4.59$ mm at wavelength $\lambda = 650$ nm.

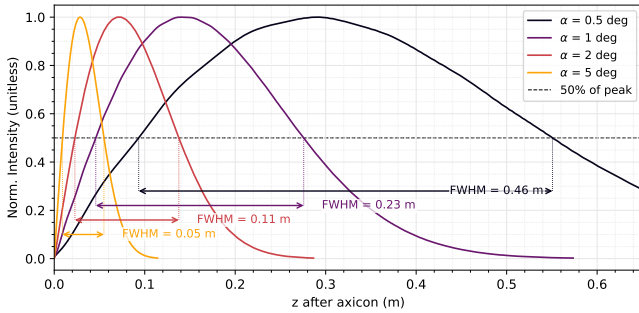


FIG. 9. QDHT model simulation of axicon angle effect on Bessel zone length, for a single axicon. The Bessel zone length is calculated from the 50% crossings of the on-axis intensity $I(0, z)$. All curves are normalized with respect to its own peak intensity, so the vertical axis should not be compared directly. See Figure 10 for comparison of on-axis intensity gain. Fixed input beam diameter $D = 4.59$ mm at wavelength $\lambda = 650$ nm.

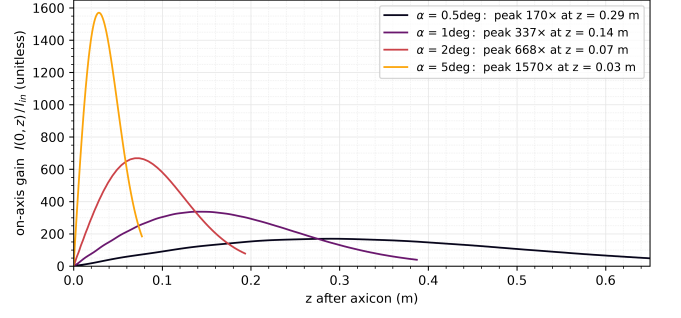


FIG. 10. QDHT model simulation of axicon angle α on on-axis intensity gain. Each curve shows the on-axis intensity $I(0, z)$ divided by the input intensity I_{in} . Gain (as a multiple of input intensity) and the peak intensity point are shown in the legend. Fixed input beam diameter $D = 4.59$ mm at wavelength $\lambda = 650$ nm.

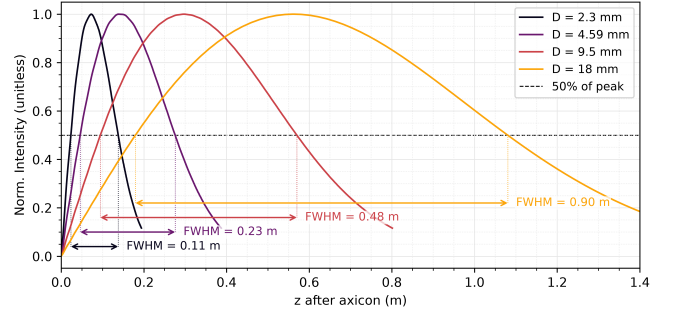


FIG. 11. QDHT model simulation of Gaussian input beam diameter D on Bessel zone length for fixed axicon angle $\alpha = 1.0^\circ$. The Bessel zone length is calculated from the 50% crossings of the on-axis intensity $I(0, z)$. All curves are normalized with respect to its own peak intensity. Fixed wavelength $\lambda = 650$ nm.

Appendix B: Axicon Alignment Tool

A command-line interface (CLI) tool was developed to assist with alignment of three-axicon system. The tool uses the CMOS camera on the gantry to continuously capture images of the beam at two planes along the axis of propagation. In most cases, the beam diameter exceeds the dimensions of the camera sensor so the tool collects a grid of images and stitches them into a single composite image.

For each composite image, the software fits a function to the beam (depending on the optic being aligned) and reports the center, radius, and eccentricity of the fitted function. Metrics from multiple planes are compared to determine the azimuthal alignment, tilt, and cone divergence/focusing of the beam (Figure 12).

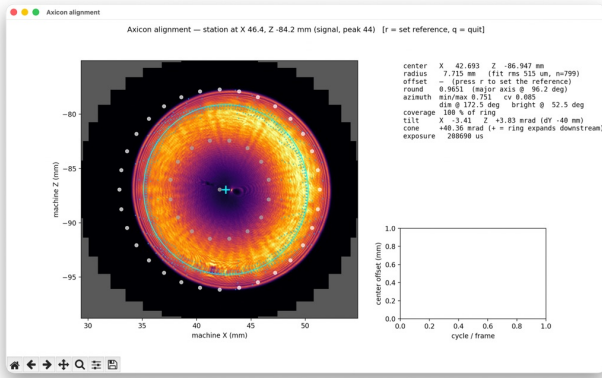


FIG. 12. Axicon 2 alignment before adjustment. The beam is not centered on the axicon, and the annulus is not circular. The inner and outer radius are drawn with dotted grey lines, with the peak intensity radial line drawn in cyan. The asymmetry of the annulus is evident in the brightness in the upper-right quadrant, compared to the lower-left quadrant.

After repeated visits to the same plane, the tool reports dX and dY offsets (Figure 13) of the beam center from previous visits. These were used to determine the direction and magnitude of adjustments needed to align the axicons (manually). A potential future direction would be to drive a motorized translation mount to automatically align the axicons.

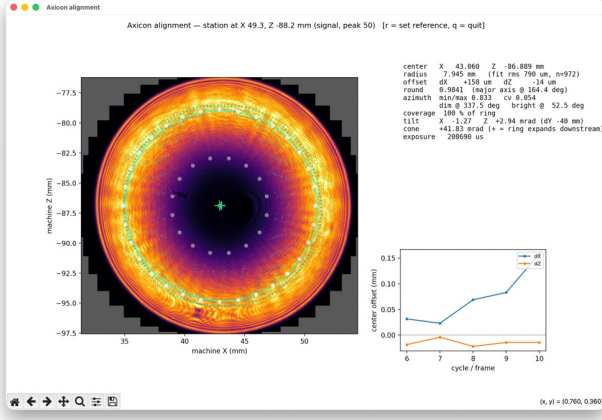


FIG. 13. Axicon 2 alignment after adjustment. The beam is centered on the axicon, and the annulus is circular. The final dX and dY offsets show alignment required $158\text{ }\mu\text{m}$ and $-12\text{ }\mu\text{m}$ adjustments in the horizontal and vertical directions respectively.

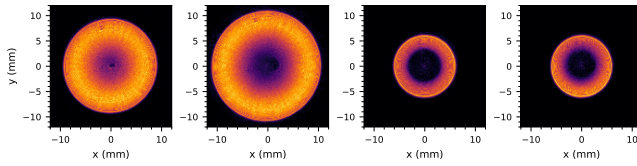


FIG. 14. (a)(b) Composite images of the annulus for candidate #2 before (a) and after (b) alignment. (c)(d) Composite images of the annulus for candidate #3 before (c) and after (d) alignment.